

# TNA Management System

## Organization-Specific AI Architecture & Implementation Plan

Angular + Django + PostgreSQL + RAG + LLM + AI Training Intelligence

### Executive Recommendation

**Do not train an LLM from scratch.** Build an organization-specific AI layer using a pretrained instruct model, retrieval-augmented generation (RAG), PostgreSQL/pgvector, TNA business data, deterministic analytics, and optional later fine-tuning.

### 1. AI Vision

The AI operates across the complete training lifecycle: employee request → approval → AI training brief → trainer materials → AI grounding → daily learning and quizzes → adaptive assessment → final assessment → competency analysis → training impact analysis → HOD/HR report → continuous learning-gap detection.

The AI is decision support. Human authorities retain control over approvals, official curriculum, permissions, and organizational decisions.

### 2. What Makes the AI Organization-Specific?

Organization-specific knowledge should come mainly from controlled sources: policies, procedures, job descriptions, competency frameworks, strategic objectives, approved training materials, previous training records, and assessment results.

Structured TNA data includes employees, departments, positions, required competencies, current competency scores, training requests, sessions, quiz attempts, final assessments, and evaluations.

### 3. Recommended Architecture

Angular → Django REST API → PostgreSQL/TNA data + AI service layer → retrieval/embeddings + LLM → grounded AI output → Angular.

AI services should cover document ingestion, retrieval, training briefs, learning assistance, quiz generation, assessment, competency analysis, training impact, organizational insights, and report generation.

### 4. Model Strategy

Start with a suitable open-source instruct model, such as a Qwen-class model, sized according to available hardware and expected usage. The exact model should be selected after measuring your CPU/GPU/RAM and concurrency requirements.

Do not pretrain a model from zero. Your project's value is the organization intelligence layer around the pretrained model.

## 5. RAG: Organization Knowledge Layer

RAG lets the AI retrieve approved organizational information at question time instead of attempting to memorize changing documents.

Document → text extraction → chunking → embeddings → pgvector → retrieval → context → LLM → grounded answer.

Example: a trainer uploads Advanced Excel training material. The system extracts and chunks it, creates embeddings, and links the knowledge to the training program, module, session, source document, and approval status.

The AI should answer from approved material and explicitly say when the required information is not available.

## 6. AI Training Lifecycle

**Request:** Employee submits topic, reason, current challenge, expected outcome, and optional competency level.

**Approval:** Department Head/Supervisor approves, rejects, or requests changes; HR/Training management controls subsequent planning.

**Training Brief:** AI proposes overview, objectives, outcomes, learning path, and duration. Trainer/HR reviews it.

**Trainer Materials:** Trainer uploads official modules, slides, notes, exercises, and references.

**Grounding:** Materials are indexed and linked to sessions so daily AI output is based on what was actually taught.

## 7. Daily AI Learning & Quiz

The AI knows the current training day, module, learning objectives, and approved material. It can generate explanations, examples, practical exercises, scenarios, and check-your-understanding activities.

Quiz questions should be grounded in the day's material and include basic, intermediate, practical, and scenario-based difficulty. Each generated question should be traceable to its source.

## 8. Adaptive Learning

Learn → Assess → Detect weakness → Targeted practice → Reassess.

If an employee repeatedly struggles with a topic such as Power Query, the system can assign targeted remediation and additional questions.

## 9. Final Assessment

After training, AI can generate a harder assessment covering the approved curriculum: knowledge, technical, practical/application, scenario-based, and organizational workflow questions.

Pre-training and post-training competency/assessment results should be compared, for example: Excel 2/5 → 4/5; Data Analysis 2/5 → 4/5; Automation 1/5 → 3/5.

## 10. Organizational Impact Analysis

AI should translate learning results into management insights such as potential reduction of manual reporting, improved accuracy, faster workflows, better dashboards, or improved productivity.

**Rule:** distinguish measured outcomes from projected benefits. Do not claim actual financial savings or productivity gains without evidence.

## 11. HOD / HR Intelligence

Reports can include employee, department, training, duration, completion, pre/post scores, competency improvement, strengths, development areas, daily performance, organizational relevance, and an AI summary.

Management questions can include: Which department has the largest competency gap? Which competencies are becoming critical? Which training produced the greatest improvement? Which employees completed training but still have significant gaps? Which skills required by strategic objectives are currently weak?

## 12. Django AI Structure

ai/ → document\_ingestion, embeddings, retrieval, training\_brief, learning\_assistant, quiz\_generator, assessment, competency\_analysis, training\_impact, organizational\_insights, report\_generator.

## 13. Core AI Entities

| Entity                             | Purpose                                                   |
|------------------------------------|-----------------------------------------------------------|
| AIRequest / AIResponse             | Track AI operations and generated outputs.                |
| KnowledgeDocument / KnowledgeChunk | Store organization/trainer sources and searchable chunks. |
| AIQuizGeneration                   | Store generated questions and source metadata.            |
| AILearningAssessment               | Store learning-performance analysis.                      |
| AICompetencyAnalysis               | Store competency-gap interpretation.                      |
| AITrainingImpactReport             | Store training impact analysis.                           |
| AIModelConfig                      | Track model/provider/version.                             |
| AIAuditLog                         | Traceability, prompts, sources, timestamps, approvals.    |

## 14. AI vs Business Logic

Django should calculate competency gaps, scores, statistics, authorization, completion, and other deterministic values. The LLM should focus on explanation, grounded generation, interpretation, question generation, summarization, and management narratives.

## 15. Security

Django authorization must decide what information an employee, supervisor, HOD, HR manager, or administrator can access. The LLM must never be the permission authority. Apply department-level filtering, document permissions, audit logs, model/version tracking, and protection against cross-employee data exposure.

## 16. Fine-Tuning

Fine-tuning is a later optimization for specialized behavior, not the initial way to load organization knowledge. After collecting sufficient high-quality approved examples, LoRA/QLoRA/PEFT can be evaluated to specialize the model.

## 17. Implementation Roadmap

1) Core TNA → 2) Training management → 3) Knowledge base/RAG → 4) AI training brief and grounded assistant → 5) Daily quizzes/adaptive assessment → 6) Competency and organizational intelligence → 7) Security/evaluation/monitoring → 8) Optional fine-tuning.

## 18. Final Architecture Decision

**Organization-Specific AI = Base LLM + Organization Knowledge (RAG) + TNA Database + Analytics + Business Rules + Optional Fine-Tuning.**

This architecture keeps organizational knowledge controllable and updateable, preserves deterministic rules in Django, supports grounded training content, enables adaptive assessment, and produces management-level learning intelligence.